

Air Pollution

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Learning Objectives

After taking this class, you will be able to:

- *acknowledge* the air pollution we are encountering
- *understand* why it is urgent to solve the air pollution problems
- *understand and illustrate* the solutions to air pollution problems

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What is air pollution?

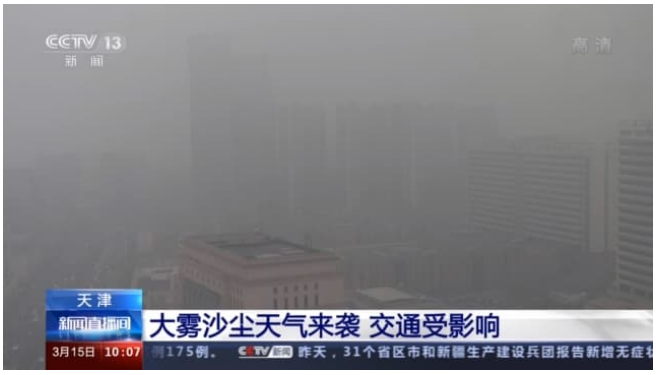
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Facts about Air Pollution in China and the World

Can see and can't see of air pollution



Beijing, March 2021



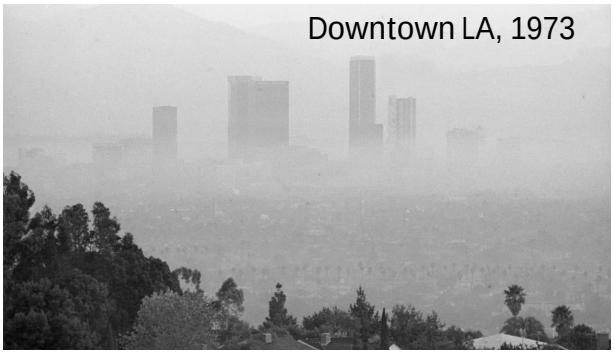
Tianjin, March 2021

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Facts about Air Pollution in China and the World

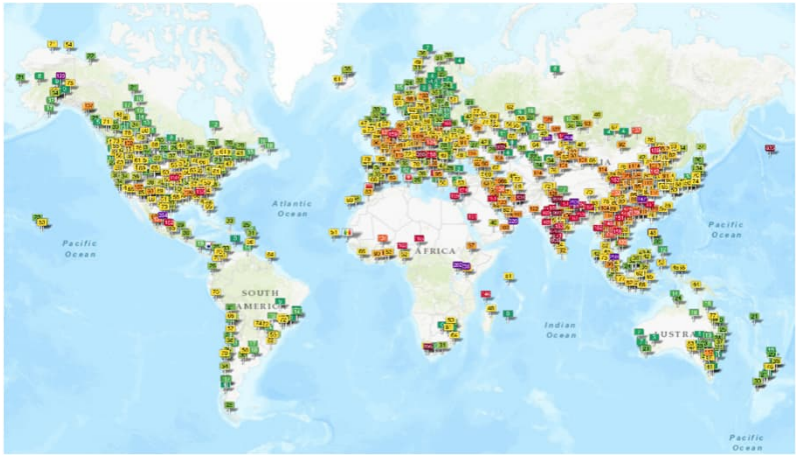
Can see and can't see of air pollution





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Facts about Air Pollution in China and the World



Real-time Air Quality Index

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Have you ever experienced air pollution?
When did you first notice air pollution issue, either in China or in your home country?

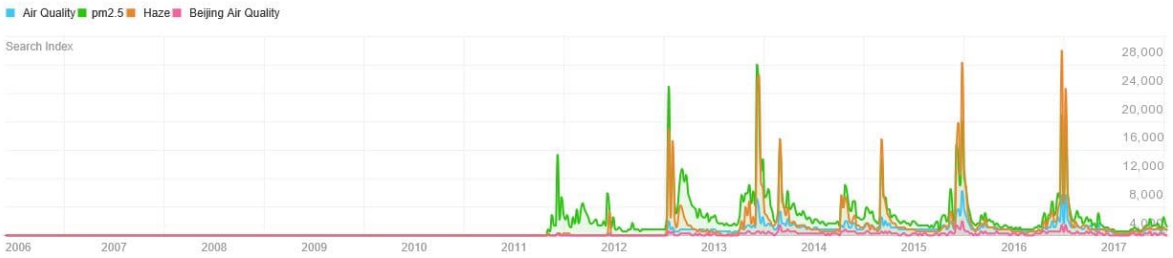
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Reaction to Air Pollution in China and the World



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Reaction to Air Pollution in China and the World



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Reaction to Air Pollution in China and the World

New Released Standards on Pollutant Concentration Limits

Pollutant	1st tier	2nd tier
SO_2	50 μg/m3	150 μg/m3
NO_2	100 μg/m3	100 μg/m3
PM10	50 μg/m3	150 μg/m3

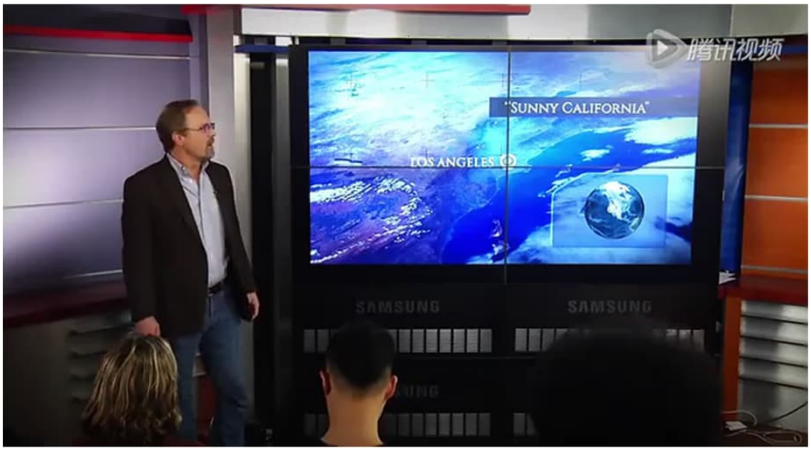
Ambient Air Quality Standards
（GB3095-1996）

Pollutant	1st tier	2nd tier
SO_2	50 μg/m3	150 μg/m3
NO_2	80 μg/m3	80 μg/m3
CO	4 mg/m3	4 mg/m3
O_3	100 μg/m3	160 μg/m3
PM10	50 μg/m3	150 μg/m3
PM2.5	35 μg/m3	75 μg/m3

Ambient Air Quality Standards
（GB3095-2012）

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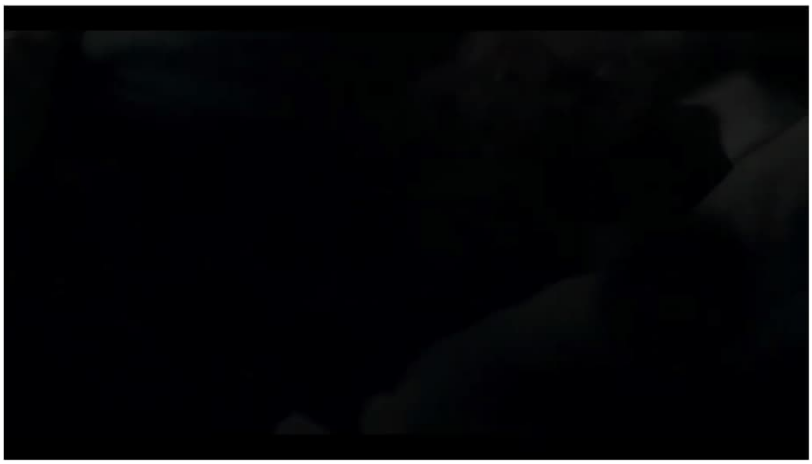
Reaction to Air Pollution in China and the World



(NASA's Goddard Space Flight Center, 2015)

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Reaction to Air Pollution in China and the World



(The Crown, Season 1 Episode 4)

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Takeaway

- Air pollution is a by-product of economic development.
- Mitigating air pollution is never an easy task. Western countries have taken decades to combat air pollution issues.
- Air pollution/smog has diverse sources. Science is essential to solve pollution issues. In fact, science is always essential for policy making. (*We will talk more about this in Lecture 8*)

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Recap and Reflection:

What are the similarities and differences between air and water pollution?

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Similarities and Differences between Air and Water Pollution

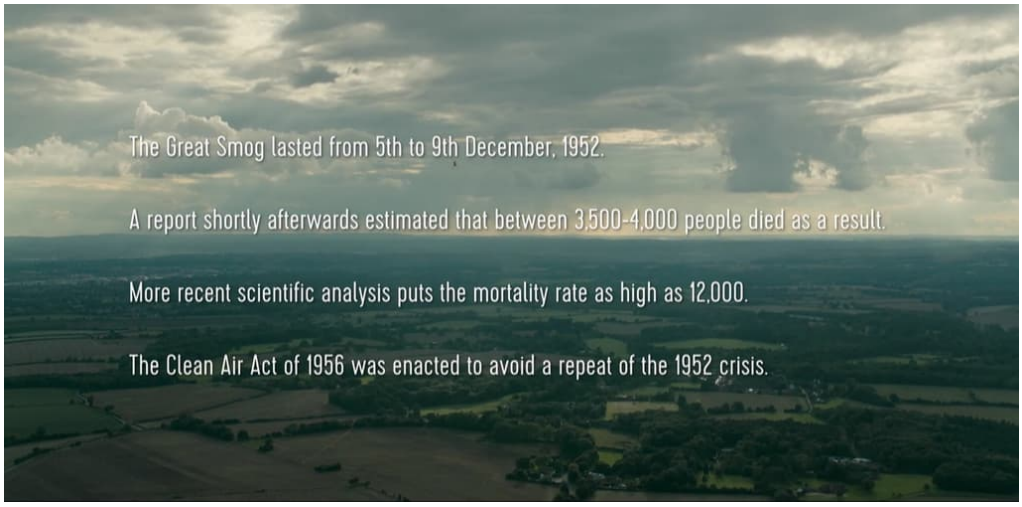
Dimensions	Air Pollution	Water Pollution
Visibility	High	Medium
Trajectory Predictability	Medium	High
Individual alternative solution	Difficult and relative high price	Medium to low price
Damage	High	High

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What are the impacts of air pollution?

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Impacts of Air Pollution



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Impacts of Air Pollution

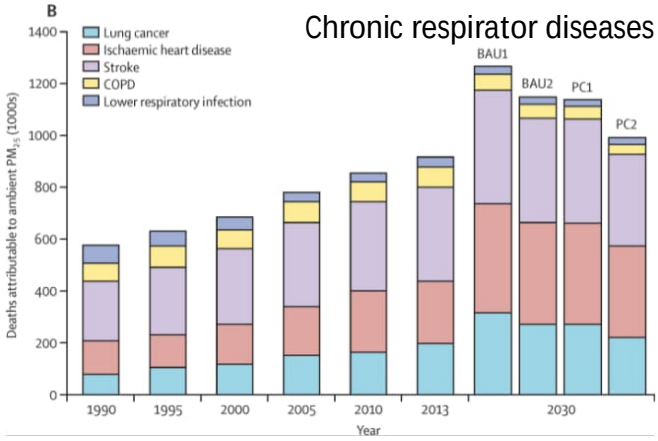
Impacts on Health I

- Polluted air was responsible in 2015 for 6.4 million deaths worldwide: 2.8 million from household air pollution and 4.2 million from ambient air pollution. (Forouzanfar et al, 2016)
- Premature death:
  - Outdoor air pollution, mostly by PM2.5, leads to average 3.3 million premature deaths per year worldwide, predominantly in Asia. (Lelieveld et al, 2015)
  - Outdoor air pollution is estimated to be responsible for five to ten percent of the total annual premature mortality in the United States. (Dedoussi et al, 2020)

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Impacts of Air Pollution

Impacts on Health II



Effect of Air Pollution Concentration on Mortality in China

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Impacts of Air Pollution

Impacts on Health III

Health benefits for air pollution control:

In 2017, as a result of substantial improvements in air quality due to the Air Pollution Prevention and Control Action Plan, there were estimated 47,240 fewer deaths and 710,020 fewer years of life lost in the 74 key cities in China than in 2013. (Huang, Pan, Guo, and Li, 2018)

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Impacts of Air Pollution

Other Economic Costs

- Medical expenditures-- an estimated US$21 billion globally in 2015
- Lost economic productivity from pollution-related disease and premature death
  - Evidence from EU (2011-2015): The estimates show that a 1μg/m3 increase in PM2.5 concentration causes a 0.8% reduction in real GDP that same year. (Dechezleprêtre, Rivers, and Stadler, 2019)
- Cost of environmental degradation

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How does smog occur?

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Factors that contribute to Air Pollution

- Many sources could contribute to smog:
  - Coal burning: coal burning power plant; boiler; civil bulk coal
  - Industrial emission
  - Motor vehicle
- Local whether and geography also matter:
  - Cities surrounded by mountain or located in a basin
  - Localities with high humidity

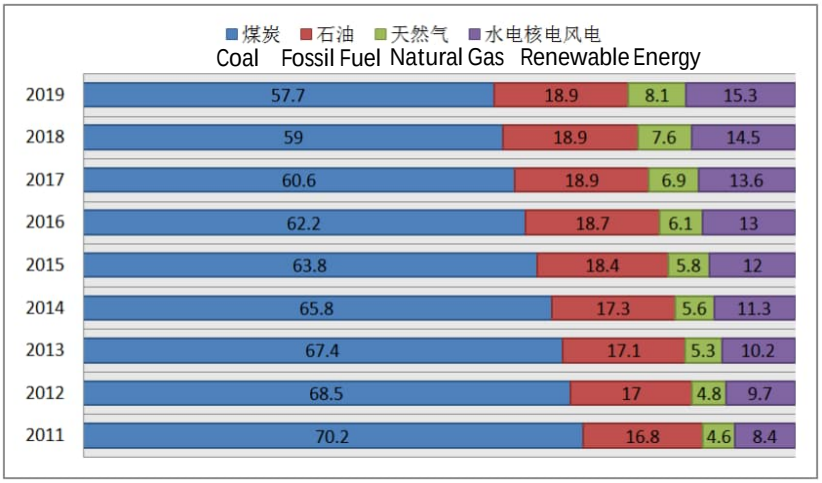
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How do we solve air pollution problems?

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Solutions to Air Pollution

Adjusting the structure of energy production and consumption



2011-2020 Energy Consumption Structure (China Energy Big Data Report, 2021)

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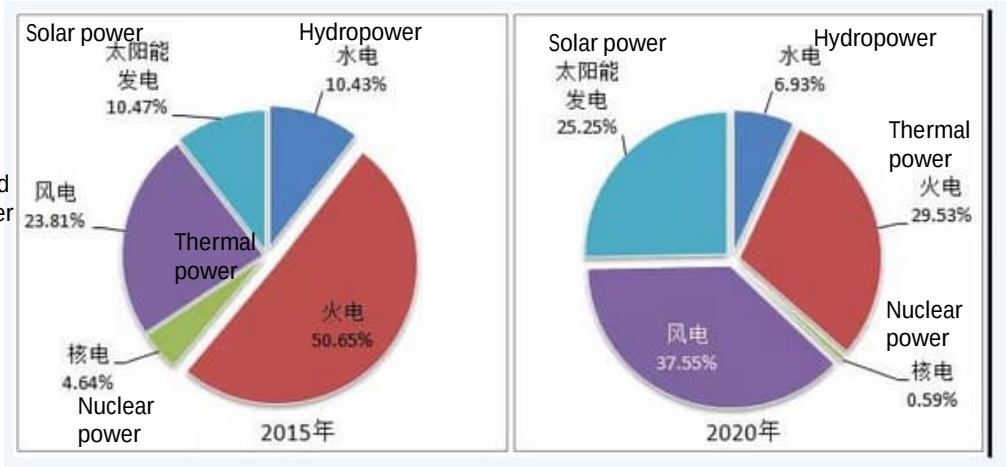
Solutions to Air Pollution



2011-2020 Structure of Installed Electric Capacity

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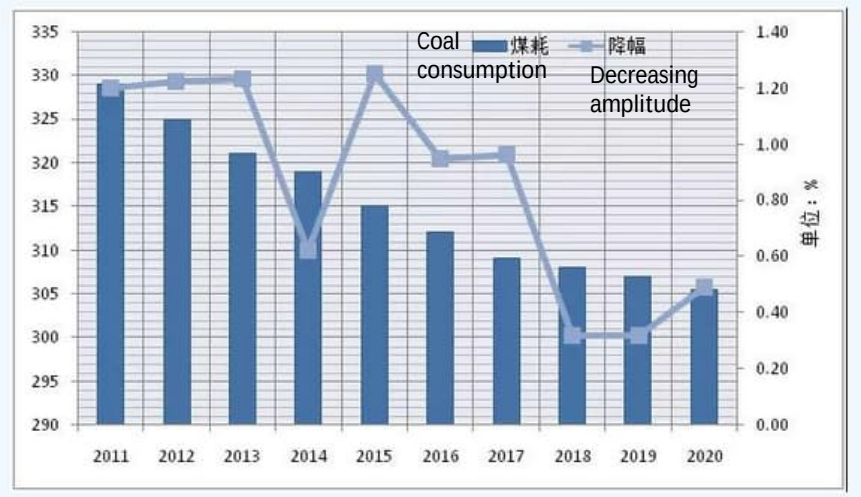
Solutions to Air Pollution



Newly installed electric capacity: Comparison between 2015 and 2020

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Solutions to Air Pollution



Coal Consumption for Power Supply (2010-2020)

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Solutions to Air Pollution

Industrial Upgrading

- Basic compliance
  - Using filters and other emission controls in production process (Installing desulfurization and denitrification equipment)
  - Recycling waste steams
- Beyond compliance
  - Environmental training for operatives and managers
  - Setting environment performance objectives as part of our annual business plan
  - Developing certifiable environmental management systems

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Pop-up Question:

Are local measures enough for air quality improvement?

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Recap: Externalities

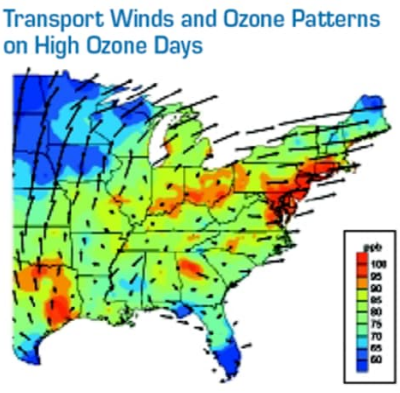
- Externalities are also referred to as spillover effects.
- It is an economic activity that imposes a negative effect on an unrelated third party. Pollution is termed an externality because it imposes costs on people who are "external" to the producer and consumer of the polluting product.



Can you provide some examples about externalities, positive or negative?

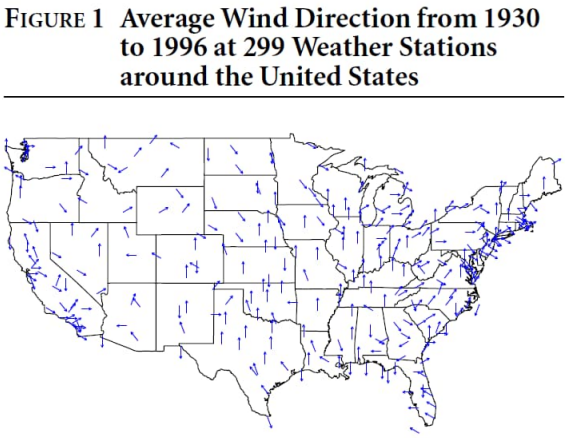
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Transboundary Air Pollution



The total pollution in any area forms from the combination of:

- local emissions of pollution, and
- upwind emissions of pollution -- pollution emitted at another location (upwind) that is then blown by wind to another location (downwind). Some upwind emissions may come from a different state.



(Monogan III, Konisky, and Woods, 2017)

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Transboundary Air Pollution

"2+26" Air Pollution Regional Joint Control Policy

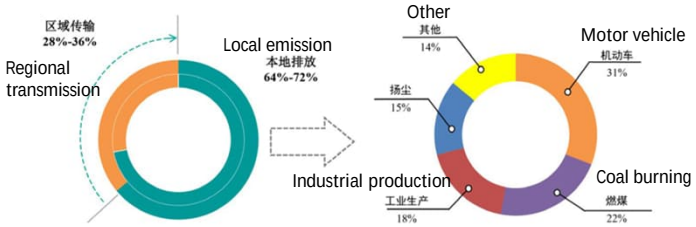
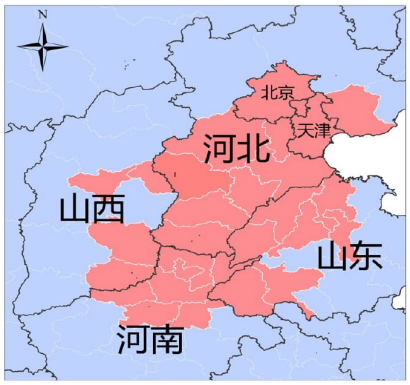


图 2.3 2013 年北京市大气中 $PM_{2.5}$ 的源解析结果

(UNEP, 2019, p.20)

- "2+26" Cities：Beijing, Tianjin, Shijiazhuang, Tangshan, Langfang, Baoding, Caozhou, Hengshui, Xingtai, Handan, Taiyuan, Yangquan, Changzhi, Jincheng, Jinan, Zibo, Jining, Dezhou, Liaocheng, Binzhou, Heze, Zhengzhou, Kaifeng, Anyang, Hebi, Xinxiang, Jiaozuo, Puyang.

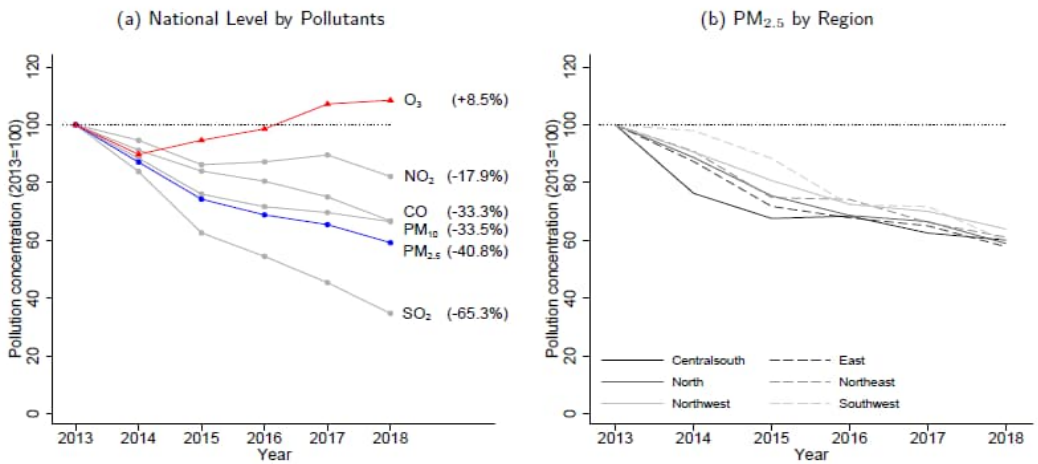
"2+26" Cities in Jing-jin-ji Area



(UNEP, 2019)

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Achievements of Air Pollution Reduction in China



Trends in Air Pollutants (Greenstone et al, 2021)

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But, it is unlikely for industries to change behavior automatically, so what should we do to induce environmental-friendly behavior?

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Achievements of Air Pollution Reduction in China

- Releasing more rigid environmental standards
  - Ambient Air Quality Standards (GB3095-2012) [环境空气质量标准]
- Intensified environmental inspection and enforcement
  - Action Plan for Prevention and Control of Air Pollution (2013-2017) [大气污染防治行动计划]
  - War on Air Pollution (2018-2020)[蓝天保卫战]
  - Central Environmental Protection Inspection [中央环保督查]

(*We will delve into the rationale and the specific practices of environmental regulation and compliances in Lecture 9.)*

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References

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